# AI/ML intern DAY 11

## Team Collaboration Task — Using TPS and Starting VTON Development

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**Date: 25-05-2026**

**Introduction**

The objective of this task was to start developing a TPS-based Virtual Try-On (VTON) workflow for intelligent garment fitting systems.

The task focused on:

- TPS transformation
- garment warping
- body alignment
- virtual cloth fitting
- VTON workflow creation

The AI system uses pose estimation and image warping techniques to align clothing images with human body posture.

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**Team Collaboration Structure**

**Team A — 2D Try-On System**

**Focus Areas**

- Clothing overlay
- TPS garment warping
- Body segmentation
- Image alignment
- Garment fitting logic
- Pose estimation

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**Team B — 3D Try-On System**

**Focus Areas**

- 3D avatars

- Body mesh generation
- Cloth simulation
- Motion tracking
- Real-time rendering

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**TPS (Thin Plate Spline) Transformation**

**Purpose**

TPS is used for:

- garment deformation
- cloth warping
- flexible body fitting
- shape alignment

TPS helps align clothing images according to:

- body posture
- shoulder width
- body orientation
- pose landmarks

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**TPS Workflow**

Human Image Input
↓
Pose Detection
↓
Body Landmark Extraction
↓
TPS Warping
↓
Garment Alignment
↓
Virtual Try-On Output

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**Starting VTON Development**

The VTON system development workflow includes:

**Step 1 — Human Image Input**

Upload:

- front-facing body image
- standing posture image
- upper body visible image

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**Step 2 — Pose Estimation**

MediaPipe Pose detects:

- shoulders
- elbows
- hips
- body structure

using AI landmark detection.

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**Step 3 — Body Segmentation**

The AI system separates:

- body regions
- background
- clothing regions

for garment fitting.

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**Step 4 — TPS Garment Warping**

The garment image is:

- stretched
- resized
- deformed
- aligned

according to body posture.

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**Step 5 — Final VTON Output**

The AI system generates:

- aligned garment fitting
- virtual try-on image
- body-cloth synchronization

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**Basic TPS-Based VTON Python Code**

Save file as:

vton_tps.py

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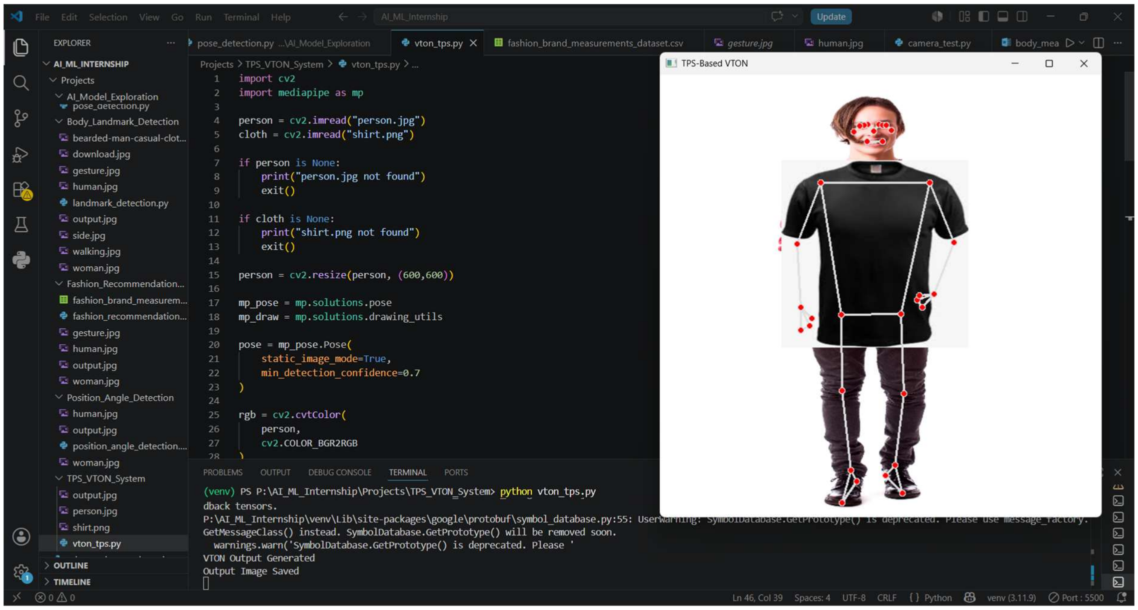
**Python Implementation**

```
import cv2
import mediapipe as mp

person = cv2.imread("person.jpg")
cloth = cv2.imread("shirt.png")

if person is None:
    print("person.jpg not found")
    exit()

if cloth is None:
    print("shirt.png not found")
    exit()

person = cv2.resize(person, (720,720))
cloth = cv2.resize(cloth, (300,300))

mp_pose = mp.solutions.pose
mp_draw = mp.solutions.drawing_utils

pose = mp_pose.Pose(
    static_image_mode=True,
    min_detection_confidence=0.7
)

rgb = cv2.cvtColor(person, cv2.COLOR_BGR2RGB)

results = pose.process(rgb)

if results.pose_landmarks:

    lm = results.pose_landmarks.landmark
```

```
    h, w, _ = person.shape

    ls = lm[11]
    rs = lm[12]

    x1, y1 = int(ls.x*w), int(ls.y*h)
    x2, y2 = int(rs.x*w), int(rs.y*h)

    shoulder_width = abs(x2-x1)

    cloth = cv2.resize(
        cloth,
        (shoulder_width+150, shoulder_width+150)
    )

    center_x = int((x1+x2)/2)

    x_offset = center_x - cloth.shape[1]//2
    y_offset = y1 - 50

    for i in range(cloth.shape[0]):
        for j in range(cloth.shape[1]):

            if (
                0 <= y_offset+i < person.shape[0]
                and
                0 <= x_offset+j < person.shape[1]
            ):

                person[
                    y_offset+i,
                    x_offset+j
                ] = cloth[i,j]

    mp_draw.draw_landmarks(
        person,
        results.pose_landmarks,
        mp_pose.POSE_CONNECTIONS
    )

    print("VTON Output Generated")

else:

    print("No Human Detected")
```

```
cv2.imwrite("output.jpg", person)

cv2.imshow("TPS-Based VTON", person)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

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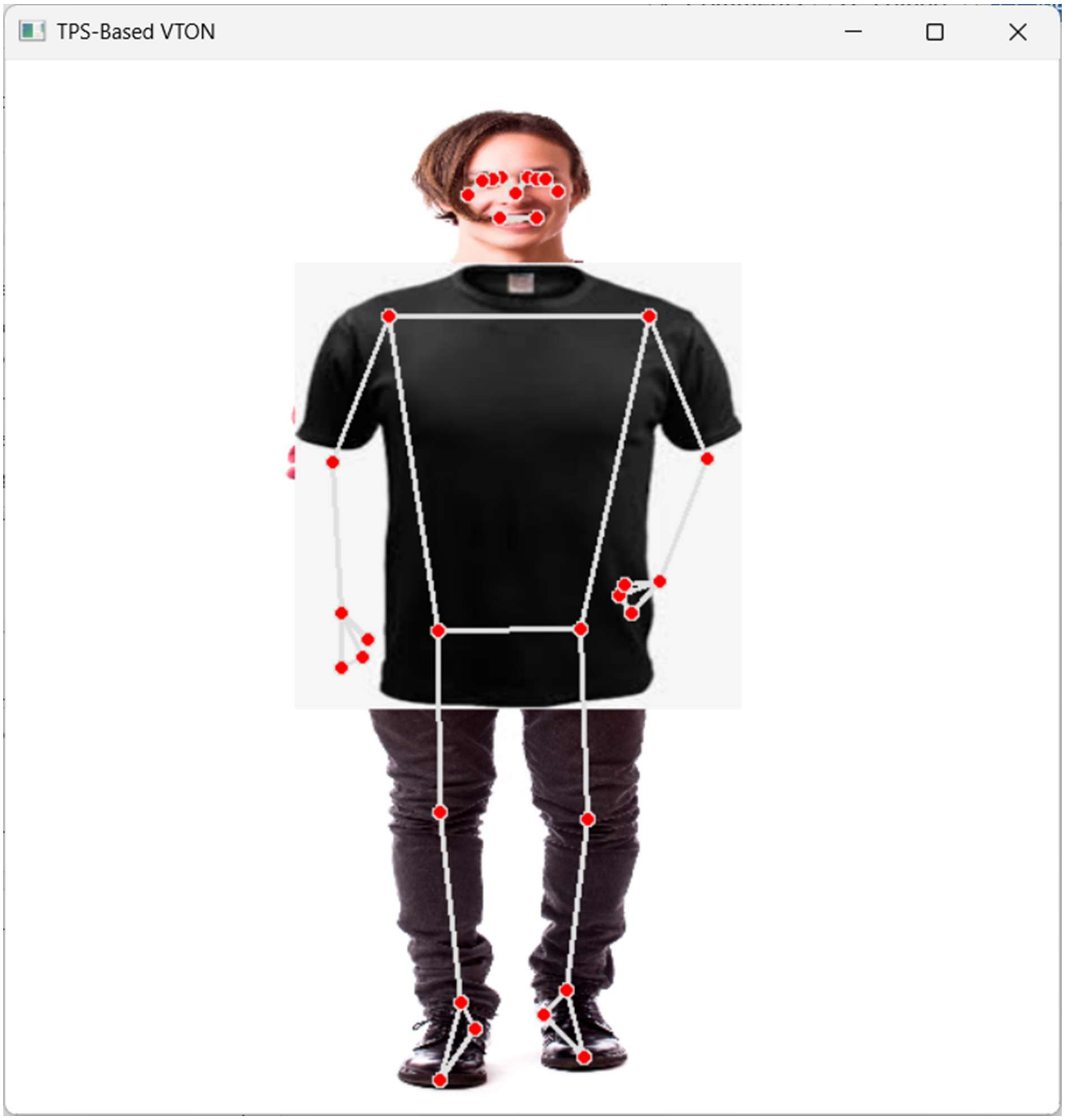
### Expected Outputs

The AI system generates:

- body landmark visualization
- TPS garment fitting
- virtual try-on output
- skeletal tracking
- aligned clothing overlay

The final image is automatically saved as:

output.jpg




## Research Findings

The research helped understand:

- TPS garment warping
- virtual cloth fitting
- body alignment systems
- VTON workflows
- garment deformation techniques

The task demonstrated how AI systems use TPS transformation and pose estimation for virtual fitting applications.

**Learning Outcomes**

During this task, the following concepts and skills were learned:

- TPS transformation
- garment warping
- virtual try-on systems
- pose estimation workflows
- body segmentation
- garment fitting logic
- AI fashion technology
- computer vision applications

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**Conclusion**

The TPS-Based Virtual Try-On (VTON) task successfully demonstrated how Artificial Intelligence and Computer Vision technologies create intelligent garment fitting systems using TPS warping and pose estimation workflows.

The task improved understanding of:

- TPS garment deformation
- VTON systems
- virtual cloth fitting
- pose alignment
- garment warping
- AI fashion technologies
- computer vision workflows